

Lecture 20: Self-gravitation and rotation

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Outline and Motivation

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- We will also incorporate the effects of the Earth's **rotation**.

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- To properly model density, we must include the effects of **self-gravitation**, which becomes important for long-period phenomena like the Earth's free oscillations.
- We will also incorporate the effects of the Earth's **rotation**.
- Finally, a scaling analysis will show us when it is valid to neglect these effects, justifying our earlier, simpler models for high-frequency seismic waves.

Including Self-Gravitation

To include gravity, we must add the gravitational potential energy to our system.

Gravitational Binding Energy

The gravitational binding energy, $\mathcal{V}_g$, is the energy required to assemble the body from matter dispersed at infinity. It can be expressed as an integral over the body's density ρ and its gravitational potential ϕ :

$$\mathcal{V}_g(t) = \frac{1}{2} \int_{M_t} \rho(\mathbf{y}, t) \phi(\mathbf{y}, t) d^3y$$

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A Non-Local Lagrangian

We add this energy to the Lagrangian. Because gravity is an "action-at-a-distance" force, the potential at one point depends on the mass distribution everywhere. This makes the Lagrangian **non-local**.

$$\mathcal{L} = \frac{1}{2} \rho ||\mathbf{v}||^2 - \underbrace{W(\mathbf{F})}_{\text{non-local}} - \frac{1}{2} \rho \zeta$$

Equations of Motion with Gravity

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Equation of Motion in an Inertial Frame

The result is an additional body force term in the equation of motion, equal to the gravitational force.

$$\rho \frac{\partial v_i}{\partial t} - \frac{\partial T_{ij}}{\partial x_j} - \rho \gamma_i = 0$$

Here, γ_i is the referential gravitational acceleration.

Working in a Co-Rotating Frame

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The Equation of Motion in a Rotating Frame

The final equation includes two new acceleration terms:

$$\rho \left(\frac{\partial \bar{v}_i}{\partial t} + \underbrace{2\epsilon_{ijk}\Omega_j \bar{v}_k}_{\text{Coriolis}} + \underbrace{\epsilon_{ijk}\epsilon_{klm}\Omega_j \Omega_l \bar{\varphi}_m}_{\text{Centrifugal}} \right) - \frac{\partial \bar{T}_{ij}}{\partial x_j} - \rho \bar{\gamma}_i = 0$$

From now on, we will work in this frame and drop the overbars for simplicity.

The Full Linearised Equations of Motion

We can now linearise the full equations about an equilibrium state of steady rotation. The resulting equation for a small displacement $\mathbf{u}$ is:

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Linearised Equation with Gravity and Rotation

$$\rho \frac{\partial^2 u_i}{\partial t^2} + 2\rho \epsilon_{ijk} \Omega_j \frac{\partial u_k}{\partial t} - \frac{\partial}{\partial x_j} (A_{ijkl} \frac{\partial u_k}{\partial x_l}) - \rho \gamma_i^1 + \dots = \text{Source}$$

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- The gravitational term γ_i^1 is an integral over the entire body, making this an **integro-partial differential equation**.

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- This equation includes inertial, Coriolis, elastic, and gravitational perturbation (γ_i^1) terms.
- The gravitational term γ_i^1 is an integral over the entire body, making this an **integro-partial differential equation**.
- Note: The equilibrium state is no longer stress-free. Centrifugal force must be balanced by internal stresses and gravity.

Scaling Analysis (1/2): When is Gravity Important?

To see when we can neglect gravity and rotation, we perform a scaling analysis, comparing the magnitude of each term. Let L be the length scale and T be the time scale of the deformation.

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Gravitational vs. Elastic Forces

$$\frac{\text{Gravitational Force}}{\text{Elastic Force}} \sim \frac{4\pi G \bar{\rho}^2 U}{\bar{\mu} U / L^2} = \frac{4\pi G \bar{\rho} L^2}{\bar{v}^2}$$

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- For planet-scale deformations where L is the Earth's radius, this ratio is of order 1. **Gravity is crucial for whole-Earth phenomena.**

Scaling Analysis (2/2): When is Rotation Important?

Coriolis vs. Inertial Term

$$\frac{\text{Coriolis Term}}{\text{Inertial Term}} \sim \frac{\bar{\rho}\Omega U/T}{\bar{\rho}U/T^2} = \Omega T$$

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Conclusion

The simpler equations of motion used in the first part of the course are valid for high-frequency seismic waves. For the long-period free oscillations of the Earth, we must include both self-gravitation and rotation.

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- Working in a **co-rotating reference frame** introduces the **Coriolis** and **centrifugal** forces.
- A ****scaling analysis**** shows that:
 - Self-gravitation is important for long-wavelength (planet-scale) deformations.
 - Rotation is important for long-period motions.
- Both effects are crucial for understanding the Earth's free oscillations but can be neglected for high-frequency body waves.